Distributed Center of Mass Estimation via Cooperative Multi-Robot Pushing

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***Abstract*—Knowledge of an object's inertial parameters is essential for effective motion planning and control in robotic manipulation tasks, yet it is commonly assumed to be known in advance. This paper proposes a distributed method for estimating the center of mass (CoM) of a planar polygonal object using a team of omnidirectional mobile robots that interact with the object solely through pushing. The robots are coordinated to generate an excitation trajectory for the object, enabling CoM estimation within a single pushing operation. The problem is decomposed into formation synthesis and force control, which jointly determine the robot contact configurations and pushing forces required to achieve the desired motion. Then, the object's CoM is estimated in a distributed manner using only local kinematic measurements, without requiring acceleration information. Finally, numerical simulations are conducted to validate the theoretical framework.**

***Index Terms*—Center of mass estimation, cooperative manipulation, formation synthesis, multi-robot systems, particle swarm optimization, pushing, recursive least squares.**

I. Introduction

Robotic manipulation [1] is a fundamental task in robotics, enabling robots to physically interact with their environment across a wide range of applications, including manufacturing, logistics, and autonomous systems. Manipulation strategies can be broadly classified into two categories: prehensile and non-prehensile manipulation, depending on the nature of the contact interactions between the robot and the manipulated object. While prehensile manipulation [2] involves establishing and maintaining a stable grasp on an object, typically through multiple contact points, non-prehensile manipulation [3], [4] achieves object motion only through contact interactions such as pushing, sliding, and rolling. Among these techniques, pushing [5] is widely employed as a simple yet effective manipulation primitive for achieving desired object poses in planar environments. Unlike grasp-based approaches, pushing does not require form or force closure, making it particularly suitable for manipulating objects that are challenging to grasp.

While single-robot pushing provides a simple and effective means of object manipulation, its capabilities may be constrained by the limited force capacity and available contact locations of an individual robot. To address these limitations, multi-robot pushing [6], [7] employs a team of robots to cooperatively manipulate an object by applying forces at different contact locations, overcoming the limitations of individual robots in terms of force capacity and manipulation flexibility. Rosenfelder et al. [8] developed a unified framework for cooperative pushing, integrating formation optimization, force allocation, and distributed control. This comprehensive approach provides a systematic solution for coordinating multiple robots to achieve reliable and efficient object transportation.

The inertial parameters of an object, particularly its CoM, play a crucial role in determining its dynamic response to applied pushing forces. While some adaptive pushing control methods do not require prior knowledge of the object's CoM [9], [10], accurate estimation of its location can provide valuable insight into the object's physical characteristics, facilitating more informed and reliable manipulation planning. In particular, CoM information can guide the determination of appropriate contact configurations and pushing strategies to produce the desired object motion. For instance, Cosgun et al. [11] employed the object's CoM as a reference for initializing gripper placement when planning linear pushing actions. Similarly, Nguyen et al. [12] introduced a distributed cascade force control strategy for cooperative object transportation, assuming that the robots are positioned on opposite sides of the projection of the object's CoM onto the contact edge.

Despite its importance in manipulation planning and control, the object's CoM is often assumed to be known or approximated from geometric information under a uniform mass distribution. However, these simplifications may not adequately capture the object's true characteristics, potentially compromising the accuracy of CoM estimates. Conventional estimation methods typically rely on dynamic models derived from the Newton–Euler equations [13]–[21], which relate applied forces and torques to the resulting linear and angular accelerations, which are susceptible to measurement noise and uncertainties that may degrade the accuracy and robustness of the estimation process. Recent studies have investigated alternative control-based approaches that enable CoM estimation through deliberately induced object motions, circumventing the need for explicit dynamic modeling [22]–[25]. However, these approaches often require multiple pushing actions and are primarily designed for single-robot manipulation, limiting their applicability to multi-robot systems. To address this problem, we propose a distributed strategy that enables a team of omnidirectional mobile robots to estimate the CoM of a planar polygonal object using only pushing contacts. The key idea is to coordinate the robots such that the net pushing force on the object vanishes while a nonzero torque is generated, inducing an approximately pure rotation about the CoM. The main contributions of this paper are as follows:

- A formation synthesis and force control framework that

generates a zero-net-force, rotation-inducing excitation for the object, with the formation optimized for robustness using ALPSO.
- A distributed RLS-based estimator that allows each robot to estimate the CoM independently from local kinematic measurements only, without requiring acceleration measurements or information exchange.
- Numerical simulations demonstrating millimeter-level estimation accuracy within a single pushing operation under measurement noise.

The remainder of this paper is organized as follows. Section II reviews related work on inertial parameter and CoM estimation. Section III formulates the problem and presents the estimation algorithm, formation synthesis, and force control. Section IV reports the simulation results, and Section V concludes the paper.

II. Related Works

To the best of our knowledge, there has been no comprehensive review dedicated specifically to center-of-mass (CoM) estimation for manipulated objects. Existing surveys primarily focus on the identification of the complete set of inertial parameters rather than CoM estimation itself. The survey by Schedlinski and Link [26] presents a literature review of existing methodologies on how to estimate these parameters in mechanical engineering, whereas that of Mavrakis and Stolkin [27] is specific to robotics. According to Mavrakis and Stolkin's taxonomy, the methods for estimating inertial parameters can be categorized into three main groups by degree of interaction — purely visual, exploratory, and fixed-object.

The first visual methods relied on the assumption of a known and uniform object density in order to simplify the estimation problem by reducing it to the computation of the object's volume. In [28], the object is represented as a collection of equally sized voxels with identical masses, from which the center of mass is determined by computing the moment of inertia matrix. One of the most widely recognized approaches is presented in [29], where the author decomposed a rigid body into polyhedral components with uniform density and computed the mass and inertia tensor of each component before combining them to obtain the inertial properties of the complete object. The computation was achieved by converting the required three-dimensional volume integrals into two-dimensional surface integrals and one-dimensional line integrals using the divergence theorem, projection, and Green's theorem. However, while visual perception can accurately capture an object's shape and volume, the underlying density distribution is generally unknown, may vary throughout the object, and cannot be directly observed solely from visual information.

Model-based approach, on the other hand, involve physical interaction with the object to perform the estimation task. These interactive techniques are commonly formulated based on the Newton–Euler equations of motion, which establish a mathematical relationship between the forces/torques (wrenches) applied to an object and its resulting motion. Properties like the center of mass (CoM), mass and the inertia tensor are treated as constant parameters within these equations and are subsequently estimated from the resulting system of equations, typically using least-squares techniques. In [13]–[15], the unknown object is rigidly connected to the robotic system during the estimation process, either through a firm grasp or by being directly attached to the end-effector. Excitation trajectories are then executed by the robotic system to retrieve the necessary data from wrist sensors. The approach proposed in [16] employed a robotic arm to perform three distinct trajectories to distinguish the individual effects of mass, CoM, and the inertia tensor. The fundamental principle of this method can also be applied to different robotic platforms. In [17], the authors used a UAV equipped with two customised grippers to grasp the object and estimated the payload mass, CoM, and inertia from the UAV dynamics and measured motion, while [18] extended this concept to a hexacopter equipped with a 2-DOF manipulator, enabling simultaneous payload estimation and adaptive motion control. Although this method can achieve the most accurate full three-dimensional inertial properties under controlled conditions, its requirements for a secure grasp and specialized force/torque sensors make it less versatile than exploratory methods in unconstrained or hazardous scenarios.

Exploratory methods exploit the measurements by utilizing more basic interaction such as poking, pushing or tilting. As mentioned, early studies of this category similarly estimate the unknown properties using a model-based approach. Yu et al. [19] employed a two-fingered robot manipulator equipped with force sensors to estimate an object's mass, 2D CoM, rotational inertia, and motion friction parameters from multiple pushing motions using a least-squares approach. For CoM estimation, they introduced C.M. lines, which are parallel to the object's straight-motion direction and pass through its CoM. These lines were identified from fingertip force and motion measurements under different pushing directions, with their intersection determining the CoM. Franchi et al. [20] developed a decentralized approach for estimating an object's mass, 2D CoM, and rotational inertia using multiple mobile robots. Information on the applied forces and contact geometry is shared among the robots through a consensus algorithm, enabling distributed estimation of the unknown inertial properties. The method was subsequently extended in [21] to rely solely on noisy end-effector velocity measurements.

Another emerging line of research focuses on CoM estimation without requiring an explicit dynamic model. A number of studies have built upon Mason's Voting Theorem [30], which establishes a geometric relationship between the pushing direction and the resulting rotational motion of a planar object, and reduce the search space through successive bisection. By additionally incorporating a modified binary search algorithm [22], Hyland et al. [23] proposed an iterative pushing method using a holonomic mobile robot and external motion capture to derive the CoM location from multiple lines on which the CoM lies, called lines of action (LoA). This work

was later extended in [24] into an infrastructure-free approach by utilizing onboard force and vision sensors. Uemura et al. [25] presented a dynamic-control-based method utilizing a nonlinear feedback control law to continuously adjust the finger's motion to suppress the object's rotation, achieving pure translational motion along the identified LoA.

A growing body of work has explored data-driven approaches that learn CoM-related information directly from observed interaction data. Gao et al. [31] introduce a force-sensor-free exploratory framework that integrates high-speed robotic pushing with Mason's Voting Theorem and recurrent neural networks to estimate the CoM of previously unseen objects with non-uniform density. In [32], the authors introduce a data-driven framework that estimates an object's 2D inertial parameters by training a Multi-Output Regression Random Forest on physical features extracted from a single robotic push. By combining analytical pushing mechanics with machine learning, the method achieves robust and accurate localization across various surface types without the need for specialized hardware. These methods enable rapid estimation with fewer physical interactions. However, their reliance on prior training data may limit their generalization to previously unseen objects.

III. Methodology

A. Formulation

The system consists of N omnidirectional mobile robots manipulating an object of polygonal shape whose motion is constrained to a 2D plane using pushing interactions, with all positions, velocities, and forces expressed in a world-fixed reference frame $\mathcal{W}$. The primary objective is for the robots to cooperatively estimate the center of mass (CoM) of the object in a distributed manner. It is assumed that the object's pose (position and orientation) and 2D geometry are known to all robots and neither the object nor the robots are subject to kinematic constraints. Each robot i has a circular footprint with radius r_R, can exert a normal force f_i within a limited range, $f_{\min} \leq f_i \leq f_{\max}$, and measure the position $\mathbf{p}_i$ and velocity $\dot{\mathbf{p}}_i$ at its contact point C_i on the object's surface, as well as the object's angular velocity ω_o. The forces applied to the object by the robots are represented by the vector $\mathbf{f} = [f_1, \ldots, f_N]^T \in \mathcal{F} := \left\{\mathbf{f} \in \mathbb{R}^N \mid f_{\min} \leq f_i \leq f_{\max},\ i = 1, \ldots, N\right\}$. The positive lower bound $f_{\min} > 0$ ensures that every robot maintains contact with the object. It is also assumed that the robots are able to maintain continuous contact with the object and accurately regulate the pushing force at their corresponding contact points throughout the manipulation process. The net pushing force and torque exerted by all robots on the object are described as:

$$\mathbf{F} = \sum_{i=1}^{N} \mathbf{F}_i \tag{1}$$

$$\tau = \sum_{i=1}^{N} (\mathbf{p}_i - \mathbf{p}_C)^{\perp T} \mathbf{F}_i \tag{2}$$

where $\mathbf{p}_C$ denotes the position of the object's CoM, and $(\cdot)^\perp$ denotes the perpendicular operator, defined as $\mathbf{v}^\perp = \mathbf{Q}\mathbf{v}$, with $\mathbf{Q} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. The force vector of robot i is $\mathbf{F}_i = f_i \mathbf{n}_i$, with $\mathbf{n}_i$ being the inward unit normal vector of the object's boundary at the contact point C_i. Friction between the object and the ground is not explicitly modeled, since an accurate dynamic prediction of the object's motion is not required. Equation (2) can be rewritten as:

$$\tau = \sum_{i=1}^{N} \mathbf{p}_i^{\perp T} \mathbf{F}_i - \mathbf{p}_C^{\perp T} \mathbf{F} \tag{3}$$

B. Estimation algorithm

When the object undergoes planar rotational motion, all contact points rotate instantaneously about the same instantaneous center of rotation (ICR). The velocity of each contact point is tangential to its instantaneous circular trajectory and therefore perpendicular to the line connecting the contact point to the ICR. When $\omega_o \neq 0$, the ICR can be geometrically determined from the velocity measurements at any contact point using the following equation:

$$\mathbf{p}_{\mathrm{ICR}} = \mathbf{p}_i + \frac{1}{\omega_o} (\dot{\mathbf{p}}_i)^\perp \tag{4}$$

The proposed distributed estimation strategy aims to generate approximately pure rotational motion of the object by coordinating the robot formation and the forces exerted by the robots such that the net force on the object vanishes, i.e., $\mathbf{F} = \mathbf{0}$, while generating a nonzero nominal desired torque τ_d. This condition decouples the net torque from the CoM position and reduces (3) to:

$$\tau = \sum_{i=1}^{N} \mathbf{p}_i^{\perp T} \mathbf{F}_i \tag{5}$$

Assuming that the object is initially at rest and that the resultant translational friction force and external disturbances are negligible, the object's CoM remains approximately stationary throughout the manipulation process, thereby approximately coinciding with the ICR. Consequently, $\mathbf{p}_C \approx \mathbf{p}_{\mathrm{ICR}}$ can be treated as a constant parameter. Multiplying (4) by ω_o yields, for robot i at time step t, the linear measurement model

$$\mathbf{y}_i(t) = \omega_o(t)\, \mathbf{p}_i(t) + \dot{\mathbf{p}}_i^{\perp}(t) = \omega_o(t)\, \mathbf{p}_C + \boldsymbol{\varepsilon}_i(t), \tag{6}$$

where $\boldsymbol{\varepsilon}_i(t)$ accounts for measurement noise and modeling errors. Each robot then locally estimates $\mathbf{p}_C$ from (6) using recursive least squares (RLS) with a forgetting factor $0 < \mu \leq 1$. Since the regressor is $\omega_o(t)\mathbf{I}_2$, the RLS estimate reduces to the weighted average

$$\hat{\mathbf{p}}_C(t) = \frac{\sum_{j=1}^{t} \mu^{t-j}\, \omega_o(j)\, \mathbf{y}_i(j)}{\sum_{j=1}^{t} \mu^{t-j}\, \omega_o(j)^2}, \tag{7}$$

in which samples with faster rotation are weighted more heavily.

The desired torque, whose magnitude should be sufficient to initiate rotational motion, can be agreed upon by all robots through communication or a consensus algorithm. Since the required torque is not known a priori, the trial can be repeated with a progressively increased τ_d, within the feasible torque range of the formation, until rotational motion is observed; if τ_d exceeds this range, the formation synthesis is repeated.

The estimation problem can now be formulated into two sub-problems: formation synthesis and force control. The objective of the formation synthesis problem is to find a formation of the robots around the object that is able to cooperatively exert the desired force and torque on the object. The force control problem determines the forces to be exerted by each robot to achieve the desired net force and torque on the object.

C. Formation synthesis

Before deriving a method for determining the robot configuration around the object, it is necessary to first define the set of feasible contact points. Since contacts are restricted to the object boundary, the contact point search space is constructed directly along its polygonal edges. A segment of length d_s adjacent to each vertex is excluded to ensure that each robot establishes a well-defined contact with a single edge. Denoting the M vertices of the polygon by $\mathbf{v}_1, \ldots, \mathbf{v}_M$ (with $\mathbf{v}_{M+1} = \mathbf{v}_1$) and the length of the k-th edge by $\ell_k = \|\mathbf{v}_{k+1} - \mathbf{v}_k\|$, assumed to satisfy $\ell_k > 2d_s$, the set of feasible contact points is given by $\mathcal{C} = \bigcup_{k=1}^{M} \left\{ \mathbf{v}_k + s\,\frac{\mathbf{v}_{k+1}-\mathbf{v}_k}{\ell_k} \;\middle|\; s \in [d_s,\, \ell_k - d_s] \right\}$. A robot in contact at $\mathbf{p}_i$ has its center at $\mathbf{r}_i = \mathbf{p}_i - r_R\mathbf{n}_i$; hence, the feasible robot center positions are obtained by offsetting the edges in $\mathcal{C}$ outward by r_R. For non-convex polygons, the offset edges adjacent to a concave vertex intersect, and a robot whose center lies beyond the intersection point would penetrate the neighboring edge. The offset edges are therefore trimmed at their mutual intersections, yielding the set of feasible robot center positions $\mathcal{R} = \{\mathbf{r} = \mathbf{p} - r_R\mathbf{n}(\mathbf{p}) \mid \mathbf{p} \in \mathcal{C},\ \|\mathbf{r} - \mathbf{q}\| \geq r_R,\ \forall\, \mathbf{q} \in \mathcal{O}\}$, where $\mathbf{n}(\mathbf{p})$ is the inward unit normal at $\mathbf{p}$ and $\mathcal{O}$ denotes the region occupied by the object. For a concave vertex with free-space angle $\alpha < \pi$, this is equivalent to replacing the margin d_s at that vertex by $\max(d_s,\, r_R \cot(\alpha/2))$, whereas no trimming is required for convex polygons. The formation synthesis is carried out directly in terms of the robot positions $\mathbf{r} = [\mathbf{r}_1^T, \ldots, \mathbf{r}_N^T]^T \in \mathbb{R}^{2N}$, and the feasible configuration space $\mathcal{P}$ is the set of all configurations $\mathbf{r}$ whose robot positions all lie in $\mathcal{R}$. The corresponding contact points are recovered as $\mathbf{p}_i = \mathbf{r}_i + r_R\mathbf{n}_i$.

Since the admissible force set $\mathcal{F}$ is closed and bounded and the wrench mapping is continuous, the corresponding manipulation space is a compact set $\mathcal{S}$ consisting of all feasible force–torque combinations, or wrenches, that can be exerted by the current formation. The manipulation space cannot be accurately determined or analyzed without prior information about the CoM position, which influences the torque generated by the applied forces. However, under the zero net force condition, this dependence vanishes, allowing the feasible torque range to be determined independently of the CoM position by solving the following linear programming problems:

$$\begin{aligned} \tau_{\max}(\mathbf{r}) &= \max_{\mathbf{f}\in\mathcal{F}} \sum_{i=1}^{N} \mathbf{p}_i^{\perp T}\mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^{N} \mathbf{F}_i = \mathbf{0}, \\ \tau_{\min}(\mathbf{r}) &= \min_{\mathbf{f}\in\mathcal{F}} \sum_{i=1}^{N} \mathbf{p}_i^{\perp T}\mathbf{F}_i \quad \text{s.t.} \quad \sum_{i=1}^{N} \mathbf{F}_i = \mathbf{0}. \end{aligned} \tag{8}$$

These two quantities define the endpoints of the interval obtained by intersecting the manipulation space with the torque axis. A formation is considered feasible if the desired torque τ_d is within the range $[\tau_{\min}(\mathbf{r}), \tau_{\max}(\mathbf{r})]$. It is also beneficial to maximize the minimum distance between the desired torque τ_d and the boundary of the feasible torque range, which can be formulated as a robustness metric J_r:

$$J_r(\mathbf{r}) = \min\left(\tau_{\max}(\mathbf{r}) - \tau_d,\ \tau_d - \tau_{\min}(\mathbf{r})\right). \tag{9}$$

The formation synthesis problem is formulated as:

$$\min_{\mathbf{r}\in\mathcal{P}} \quad J(\mathbf{r}) = -J_r(\mathbf{r}) \tag{10a}$$

$$\text{s.t.} \quad J_r(\mathbf{r}) \geq 0, \tag{10b}$$

$$\|\mathbf{r}_i - \mathbf{r}_j\| \geq d_{\min}, \qquad \forall i \neq j. \tag{10c}$$

where $d_{\min} > 2r_R$ is the minimum Euclidean distance between the robots to avoid collision. Constraint (10b) is equivalent to $\tau_d \in [\tau_{\min}(\mathbf{r}), \tau_{\max}(\mathbf{r})]$, i.e., there exists an admissible force vector $\mathbf{f} \in \mathcal{F}$ that produces zero net force and the desired torque τ_d. It is essential that the constraints (10b) and (10c) are satisfied to ensure that the desired wrench can be exerted by the formation without violating physical feasibility, while minimizing the objective function yields a configuration that is optimal with respect to the selected criteria. The objective function (10a) encodes a desirable property, which can be modified to include other metrics. In this case, only rotational robustness is considered. The formation synthesis problem is a non-convex optimization problem, which can be tackled using metaheuristic optimization methods such as particle swarm optimization (PSO). To maintain the distributed nature of the system, each robot can run the optimization algorithm independently and share its best particle with its neighbors. Once the optimal formation has been determined, the robots autonomously negotiate which position each robot should occupy. Based on the resulting assignment, each robot then navigates toward its designated position and establishes contact with the object boundary, thereby allowing the team to establish the desired manipulation configuration in a distributed manner.

D. Force control

The next problem to be solved is to determine the forces to be exerted by each robot to achieve the desired net force and torque on the object. While a straightforward approach would be to solve (1) and (5) using a least-squares approach, this would fail to account for the physical limitations of the forces exerted by the robots, potentially yielding a physically infeasible solution. To address this limitation, the force control problem is formulated as an optimization problem:

$$\min_{\mathbf{f}} \quad \left(\sum_{i=1}^{N} \mathbf{p}_i^{\perp T}\mathbf{F}_i - \tau_d\right)^2 + \lambda\|\mathbf{f}\|_2^2, \qquad \lambda > 0 \tag{11a}$$

$$\text{s.t.} \quad \mathbf{F} = \mathbf{0}, \tag{11b}$$

$$f_{\min} \le f_i \le f_{\max}, \qquad i = 1, \ldots, N. \tag{11c}$$

Satisfying constraints (11b) and (11c) is essential to ensure force balance and compliance with actuator limits, while the objective function encourages the generation of the desired rotational torque. It is desirable that the net torque be as close as possible to the desired torque τ_d. However, it is not required to exactly match τ_d since only a torque sufficient to induce rotational motion is required for the estimation task. The second term in the objective function (11a) is a quadratic regularization term that penalizes large force magnitudes, which is beneficial for energy efficiency and avoiding actuator saturation. The regularization coefficient λ is chosen to be sufficiently small to prioritize torque tracking while discouraging unnecessarily large pushing forces. The optimization problem can be solved using standard convex optimization techniques, such as quadratic programming. The uniqueness of the optimal solution can be established by examining the convexity of the objective function. Let $\mathbf{a} = [a_1, \ldots, a_N]^T$, where $a_i = \mathbf{p}_i^{\perp T}\mathbf{n}_i$. The Hessian matrix of the objective function (11a) is given by $\mathbf{H} = 2\mathbf{a}\mathbf{a}^T + 2\lambda\mathbf{I}$. Since $\mathbf{a}\mathbf{a}^T$ is positive semidefinite and $\lambda > 0$, the Hessian is positive definite, implying that the objective function is strongly convex. Furthermore, the zero net force constraint and actuator limits define a compact and convex feasible set, which is nonempty for any formation satisfying constraint (10b), since the formation synthesis computes $\tau_{\min}$ and $\tau_{\max}$ over the same set $\mathcal{F}$. Consequently, the optimization problem admits a unique global optimal solution. Moreover, since the problem is a small convex quadratic program with only N decision variables, two equality constraints, and simple bound constraints, it can be solved efficiently by off-the-shelf QP solvers, such as OSQP, qpOASES, or MATLAB's `quadprog`, typically within milliseconds. This allows the forces to be recomputed at every control step as the contact points move with the rotating object. Each robot can independently solve this optimization problem using the formation obtained above, ensuring a distributed implementation of the force control strategy.

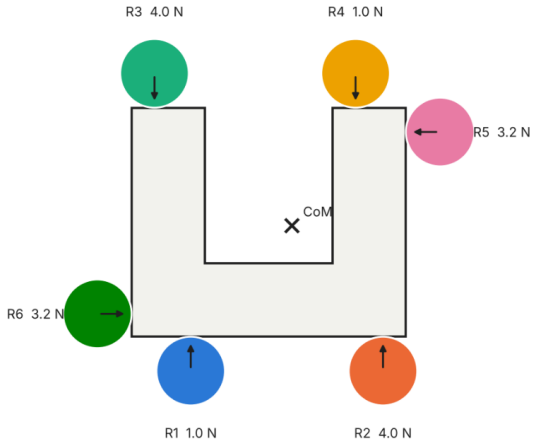


Fig. 1. Optimal formation of $N = 6$ robots around the U-shaped object obtained by ALPSO, with the pushing force of each robot computed by the force control problem. The arrows indicate the force directions, and the cross marks the true CoM.

IV. Results

In this simulation experiment, we consider a team of $N = 6$ robots with radius $r_R = 0.15\,\text{m}$ and pushing force bounds $f_{\min} = 1\,\text{N}$ and $f_{\max} = 10\,\text{N}$, which cooperatively perform an estimation task to obtain the CoM of a U-shaped object. The object has a mass of $m = 3\,\text{kg}$ and a moment of inertia $I = 0.1\,\text{kg}{\cdot}\text{m}^2$. First, the formation synthesis problem is addressed using ALPSO to determine the optimal robot configuration. The obtained formation is illustrated in Fig. 1 and achieves a robustness of $J_r = 5.66\,\text{N}{\cdot}\text{m}$. After the robots move to their designated positions, each robot independently solves the quadratic programming problem to determine its pushing force while maintaining zero resultant force and generating a torque close to the desired value τ_d. The resulting pushing forces are also annotated in Fig. 1. The pushing forces of robots R1 to R6 are 1.0, 4.0, 4.0, 1.0, 3.2, and 3.2 N, respectively, yielding zero net force and a torque of $2.93\,\text{N}{\cdot}\text{m}$.

During the rotational motion, each robot independently estimates the object's CoM using RLS, based on successive measurements of its contact point position and velocity and the object's angular velocity. Note that, since the contact point is assumed to be fixed relative to the robot, its position and velocity can be derived from the robot's pose, linear velocity, and angular velocity. To account for sensing imperfections, the contact point position and velocity and the object's angular velocity measured by each robot are corrupted by independent zero-mean Gaussian white noise with standard deviations $\sigma_p = 2\,\text{mm}$, $\sigma_v = 0.01\,\text{m/s}$ and $\sigma_\omega = 0.02\,\text{rad/s}$, respectively.

The estimation errors of all robots are shown in Fig. 2. Large errors are observed initially, since few measurements are available and the angular velocity of the object is still small at the onset of rotation. As more measurements are collected, the estimates of all robots converge rapidly, with the errors falling below 1 cm after approximately 0.6 s. After 1.5 s, the

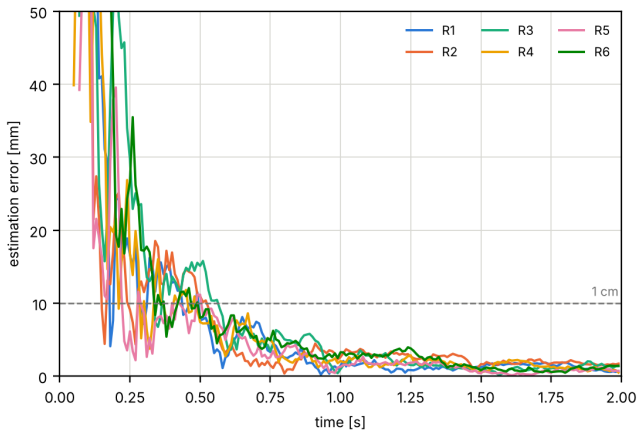


Fig. 2. CoM estimation error of each robot over time. The dashed line indicates an error of 1 cm.

estimation errors of all robots remain below 3 mm and settle at around 1 mm by the end of the experiment, which is smaller than the standard deviation of the position measurement noise owing to the averaging effect of RLS. Moreover, the estimates of the six robots converge to nearly the same value, showing that each robot can independently obtain an accurate CoM estimate without sharing its measurements.

V. Conclusion

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